有理函数 $\frac{P(x)}{Q(x)}$ 的拆分.

若 $Q(x)=(x-a_1)^{n_1}\cdots(x-a_k)^{n_k}\cdot(x^2+p_1x+q_1)^{m_1}\cdots(x^2+p_lx+q_l)^{m_l}$

这里 $x^2+p_ix+q_i=(x+\lambda_i)^2+c_i^2$ $(c_i>0)$

则 $\frac{P(x)}{Q(x)}=A_0+A_1x+\cdots+A_nx^n$

$+\frac{A_{11}}{x-a_1}+\cdots+\frac{A_{1n_1}}{(x-a_1)^{n_1}}+\cdots+\frac{A_{k1}}{x-a_k}+$

$\cdots+\frac{A_{kn_k}}{(x-a_k)^{n_k}}+\cdots$

$+\frac{B_{11}x+D_{11}}{x^2+p_1x+q_1}+\cdots+\frac{B_{1m_1}x+D_{1m_1}}{(x^2+p_1x+q_1)^{m_1}}$

$+\cdots\frac{B_{l1}x+D_{l1}}{x^2+p_lx+q_l}+\cdots+\frac{B_{1m_l}+D_{lm_l}}{(x^2+p_lx+q_1)^{m_l}}$

(1) $\int \frac{dx}{x-a_i} = \ln|x-a_i| + C$

(2) $\int \frac{dx}{(x-a_i)^k} = -\frac{1}{k-1} \cdot \frac{1}{(x-a_i)^{k-1}} + C$

$(k>1,\ k \in \mathbb{Z}^+)$

(3) $\int \frac{dx}{x^2+p_i x+q_i} = \int \frac{dx}{(x+\lambda_i)^2+c_i^2}$

$= \frac{1}{c_i} \arctan \frac{x+\lambda_i}{c_i} + C$

(4) $\int \frac{dx}{(x^2+p_i x+q_i)^k} = \int \frac{dx}{((x+\lambda_i)^2+c_i^2)^k}$

令 $u = x+\lambda_i$，则上式为

$\int \frac{du}{(u^2+c_i^2)^k}$，见上一节例 2.15.

通过递推可求.

(5). $\int \frac{x+\lambda_i}{(x^2+p_i x+q_i)^k}\,dx \quad (k>1,\ k\in\mathbb{Z}^+)$

$$= \int \frac{(x+\lambda_i)\,dx}{\left((x+\lambda_i)^2+c_i^2\right)^k}$$

$$= \frac{1}{2}\int \frac{d\left((x+\lambda_i)^2+c_i^2\right)}{\left((x+\lambda_i)^2+c_i^2\right)^k}$$

$$= -\frac{1}{2(k-1)}\cdot\frac{1}{\left((x+\lambda_i)^2+c_i^2\right)^{k-1}}+C$$

(注: $Ax+D = A(x+\lambda_i)+(D-A\lambda_i)$)

因此，我们可求出 $\int \frac{P(x)}{Q(x)}\,dx$

- 求不定积分

$$\int \frac{x-2}{x^2-8x+15}\,dx$$

分析：$x^2-8x+15=(x-3)(x-5)$.

令 $\frac{x-2}{x^2-8x+15}=\frac{A}{x-3}+\frac{B}{x-5}$

$$A(x-5)+B(x-3)=x-2$$

待定系数，解得 $\begin{cases} A=-\frac{1}{2} \\ B=\frac{3}{2} \end{cases}$

故 $\int \frac{x-2}{x^2-8x+15}\,dx=-\frac{1}{2}\int \frac{dx}{x-3}+$

$\frac{3}{2}\int \frac{dx}{x-5}=-\frac{1}{2}\ln|x-3|+\frac{3}{2}\ln|x-5|+C$

例：$\int \frac{x^5+x^4-8}{x^3-x}\,dx$

解：$\frac{x^5+x^4-8}{x^3-x} = x^2+x+1+\frac{x^2+x-8}{x^3-x}$

令 $\frac{x^2+x-8}{x^3-x} = \frac{A}{x}+\frac{B}{x-1}+\frac{C}{x+1}$.

则 $A(x-1)(x+1)+Bx(x+1)+Cx(x-1)$

$= x^2+x-8$

待定系数，解得

$A=8,\ B=-3,\ C=-4$

故 $\int \frac{x^5+x^4-8}{x^3-x}\,dx$

$= \int \left((x^2+x+1)+\frac{8}{x}-\frac{3}{x-1}-\frac{4}{x+1}\right)dx$

$$= \frac{1}{3}x^3 + \frac{1}{2}x^2 + x + 8\ln|x| - 3\ln|x-1|$$

$$- 4\ln|x+1| + C$$

- 复杂一点的有理函数积分：

例：$\int \frac{2x+2}{(x-1)(x^2+1)^2}\,dx$

解：设 $\frac{2x+2}{(x-1)(x^2+1)^2} = \frac{A}{x-1} + \frac{Bx+C}{x^2+1} + \frac{Dx+E}{(x^2+1)^2}$

则 $A(x+1)^2 + (Bx+C)(x^2+1)(x-1) + (Dx+E)(x-1)$

$$= 2x+2$$

令 $x=1$，得 $A=1$.

令 $x=i$，得 $(Di+E)(i-1) = 2i+2$

$$Di+E = \frac{2i+2}{i-1} = -2i$$

故 $D=-2$. $E=0$.

将 $A=1, D=-2, E=0$ 代入原式，令

$x=0$，得 $1+C\cdot(-1)+0=2$

故 $C=-1$

再令 $x=-1$，得 $4+(-B-1)\cdot(-4)-4=0$

故 $B=-1$.

于是 $$\frac{2x+2}{(x-1)(x^2+1)^2}=\frac{1}{x-1}-\frac{x+1}{x^2+1}-\frac{2x}{(x^2+1)^2}$$

$$\int\frac{2x+2}{(x-1)(x^2+1)^2}dx=\int\frac{dx}{x-1}-\int\frac{(x+1)\,dx}{x^2+1}$$

$$-\int\frac{2x}{(x^2+1)^2}dx=\ln|x-1|-\frac{1}{2}\ln|x^2+1|$$

$$-\arctan x+\frac{1}{x^2+1}+C$$

- 例13.4见书

- 例 $\int \frac{x^2+1}{(x^2-2x+2)^2}dx$

解: $\frac{x^2+1}{(x^2-2x+2)^2} = \frac{(x^2-2x+2)+(2x-1)}{(x^2-2x+2)^2}$

$= \frac{1}{x^2-2x+2} + \frac{2x-1}{(x^2-2x+2)^2}$

$\int \frac{dx}{x^2-2x+2} = \int \frac{dx}{(x-1)^2+1} = \arctan(x-1) + C_1$

$\int \frac{2x-1}{(x^2-2x+2)^2}dx = \int \frac{2(x-1)+1}{((x-1)^2+1)^2}dx$

$= \int \frac{2(x-1)dx}{((x-1)^2+1)^2} + \int \frac{dx}{((x-1)^2+1)^2}$ (*)

令 $t=x-1$, 则

上式 $= \int \frac{2t\,dt}{(t^2+1)^2} + \int \frac{dt}{(t^2+1)^2}$

其中 $\int \frac{2t\,dt}{(t^2+1)^2} = -\frac{1}{t^2+1} + C_2$

$$\int \frac{dt}{(t^2+1)^2} = \frac{t}{2(t^2+1)} - \frac{1}{2}\arctan t + C_3$$

(实际上: $\int \frac{dt}{t^2+1} = \frac{t}{t^2+1} + \int t \cdot \frac{2t\,dt}{(t^2+1)^2}$

$$= \frac{t}{t^2+1} + 2\int \frac{dt}{t^2+1} - 2\int \frac{dt}{(t^2+1)^2}$$)

从而 (* 式) $= \frac{t-2}{2(t^2+1)} + \frac{1}{2}\arctan t + C_4$

$$= \frac{x-3}{2(x^2-2x+2)} + \frac{1}{2}\arctan(x-1) + C_4$$

原式 $= \frac{x-3}{2(x^2-2x+2)} + \frac{3}{2}\arctan(x-1) + C$

作业：

习题4-3A类

(2)　　(4)(5)(6)

习题4-3B类

(1)、(4)